An ideal gas obeys the equation PV = NkT at thermodynamic equilibrium. Since all four macroscopic variables (P,V,T and N) are connected by this equation, we cannot change one variable alone. For example, if we push the piston of a gas container, the volume of the gas will decrease but pressure will increase or if heat is supplied to the gas, its temperature will increase, pressure and volume of the gas may also increase.

There is another example of equation of state called van der Waals equation. Real gases obey this equation at thermodynamic equilibrium. The air molecules in the room truly obey van der Waals equation of state. But at room temperature with low density we can approximate it into an ideal gas.

8.5 ZEROTH LAW OF THERMODYNAMICS

The zeroth law of thermodynamics states that if two systems, *A* and *B*, are in thermal equilibrium with a third system, *C*, then *A* and *B* are in thermal equilibrium with each other.

Consider three systems A, B and C which are initially at different temperatures. Assume that A and B are not in thermal contact with each other as shown in Figure 8.18 (a) but each of them is in thermal contact with a third system C. After a lapse of time, system A will be in thermal equilibrium with C and B also will be in thermal equilibrium with C. In this condition, if the systems A and B are kept in thermal contact as shown in the Figure 8.18 (b), there is no flow of heat between the systems A and B. It implies that the system A and B are also in thermal equilibrium with each other. Once the three systems are at thermal equilibrium, there will be no heat flow between them as they are at the same temperature. This can be mathematically expressed as if, $T_A = T_C$ and $T_B = T_C$, it implies that $T_A = T_B$, where T_A, T_B and T_C are the temperatures of the systems A, B, and C respectively.

Temperature is the property which determines whether the system is in

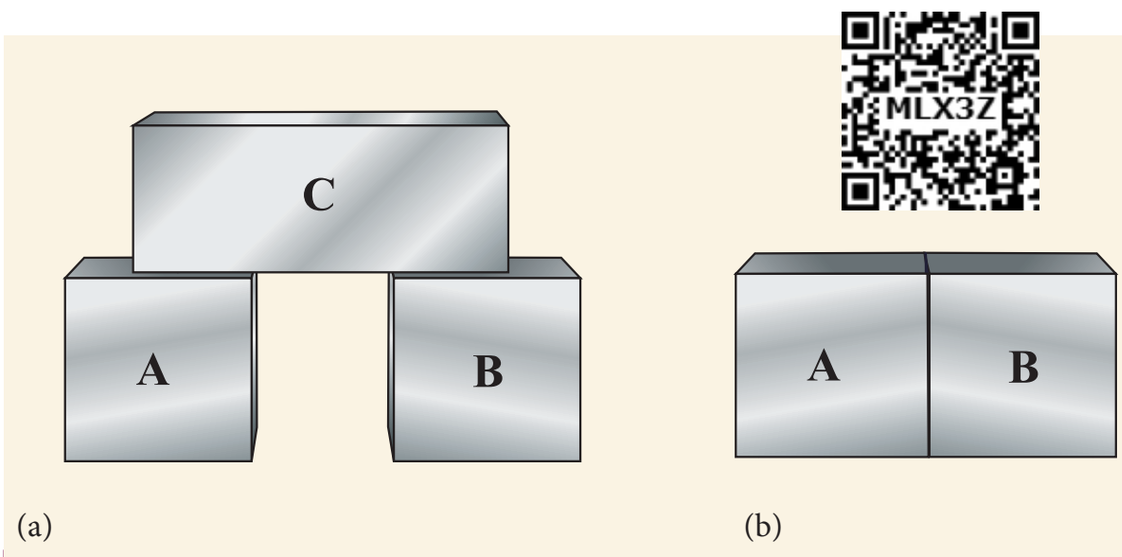


Figure 8.18 (a) Two systems A and B in thermal contact with object C separately (b) If systems A and B are in thermal contact, they are also in thermal equilibrium with each other.

UNIT-8(XI-Physics_Vol-2).indd 115 02-01-2023 15:25:43